

FTP COMMANDS & SECURITY LAB CHEATSHEET

1. LAB ENVIRONMENT OVERVIEW

Target Server	127.0.0.1 (Docker Container: ftp-lab-server)
Control Port	21 (TCP)
Data Ports	20 (Active Mode), 30000-30009 (Passive Mode Range)
Anonymous Login	Username: anonymous Password: (any) Root: /var/ftp/anon
Standard User	Username: ftpuser Password: password123 Root: /home/ftpuser
Admin User	Username: admin Password: adminpass Root: /home/admin

2. INTERACTIVE FTP CLI COMMANDS (FTP 127.0.0.1)

passive	Enable Passive Mode (REQUIRED in Docker to avoid 500 Illegal PORT)
ls / dir	List files and directories in current remote folder
cd <directory>	Change remote working directory
cdup	Change remote directory to parent folder
pwd	Print current remote working directory
get <remote> [local]	Download a file from server to local machine
put <local> [remote]	Upload a file from local machine to server
mget *.txt	Download multiple files matching wildcard pattern
mput *.txt	Upload multiple files matching wildcard pattern
delete <filename>	Delete specified file on remote server
mkdir / rmdir	Create or remove directory on remote server
binary / ascii	Set transfer mode (binary for images/executables, ascii for text)
bye / quit / exit	Close connection and exit FTP client

3. CURL COMMANDS (QUICK SCRIPTING & AUTOMATION)

curl ftp://127.0.0.1/pub/	List directory contents anonymously
curl ftp://127.0.0.1/pub/file.txt	Download file anonymously

<code>curl -T file.txt ftp://127.0.0.1/incoming/</code>	Upload file anonymously to /incoming
<code>curl -u user:pass ftp://127.0.0.1/file.txt</code>	Authenticated file download
<code>curl -u user:pass -T local.txt ftp://127.0.0.1/</code>	Authenticated file upload

4. RAW PROTOCOL COMMANDS (NETCAT/TELNET ON PORT 21)

USER <username>	Send username (Server returns 331 Please specify password)
PASS <password>	Send password (Server returns 230 Login successful or 530)
PWD	Query current working directory
CWD <dir>	Change working directory
PASV	Request passive data port (Returns 227 Entering Passive Mode)
PORT a,b,c,d,p1,p2	Specify client active data port
LIST	Request directory list over open data channel
RETR <filename>	Retrieve/download file over data channel
STOR <filename>	Store/upload file over data channel
QUIT	Terminate control session

5. SECURITY AUDITING & DEFENSIVE COMMANDS

<code>nmap -p 21 -sV --script ftp-* 127.0.0.1</code>	Audit FTP service, anonymous access, and server banners
<code>hydra -l ftpuser -P wordlist.txt ftp://127.0.0.1</code>	Perform single-user dictionary brute-force test
<code>hydra -L users.txt -P pass.txt -t 16 ftp://127.0.0.1</code>	Perform multi-user parallel brute-force test
<code>sudo tcpdump -i lo port 21 -A</code>	Sniff unencrypted plaintext USER and PASS packets
<code>docker exec -it ftp-lab-server tail -f /var/log/vsftpd.log</code>	Monitor live vsftpd authentication logs
<code>python3 brute_force_lab.py</code>	Run Python demonstration brute-force script

6. HYDRA COMMANDS & FLAGS

-L <file>	Path to candidate username list file.
-P <file>	Path to candidate password list file.
-f	Finish/exit immediately when a valid credential pair is discovered.
-o <file>	Output file path to log discovered credentials.
-I	Ignore existing session restore files (hydra.restore).
-vv	Double verbose mode; outputs each attempt and detailed progress.
-t <threads>	Sets parallel thread/connection count (default: 16).